

DataPLANT Tools & Services

And how they support community RDM

Dominik Brillhaus – CEPLAS, HHU Düsseldorf

FDM-Werkstatt 2026

March 24th, 2026



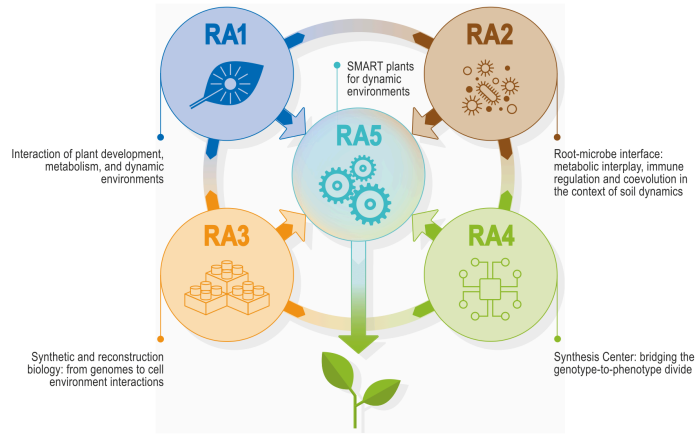
CEPLAS

Cluster of Excellence on Plant Sciences

Data))((PLANT

Cluster of Excellence on Plant Sciences (CEPLAS)

SMART Plants in Dynamic Environments

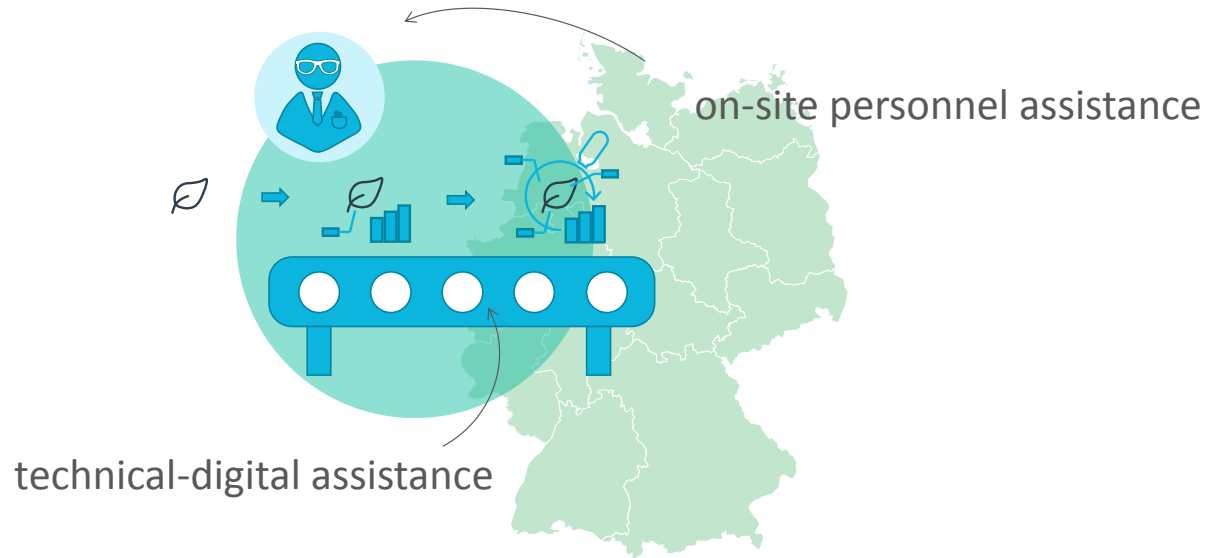


UNIVERSITÄT
ZU KÖLN



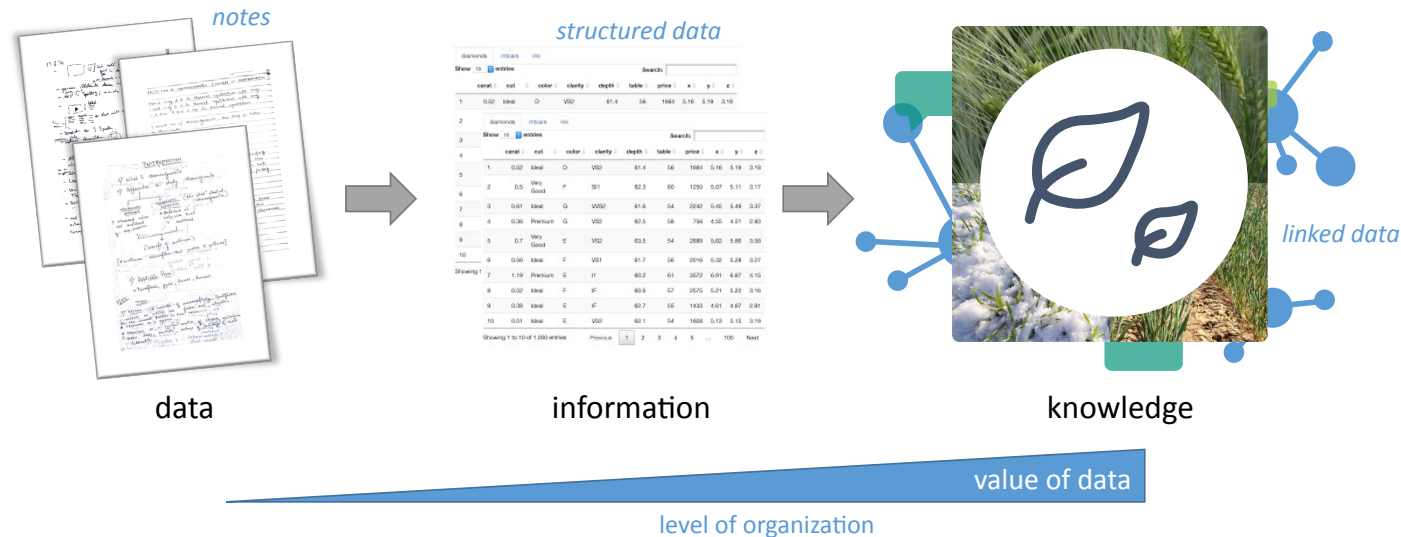
- ~50 groups
- Grad School
- Technology platforms
 - Metabolomics
 - Imaging
 - Data science and management

DataPLANT – *NFDI4plants*



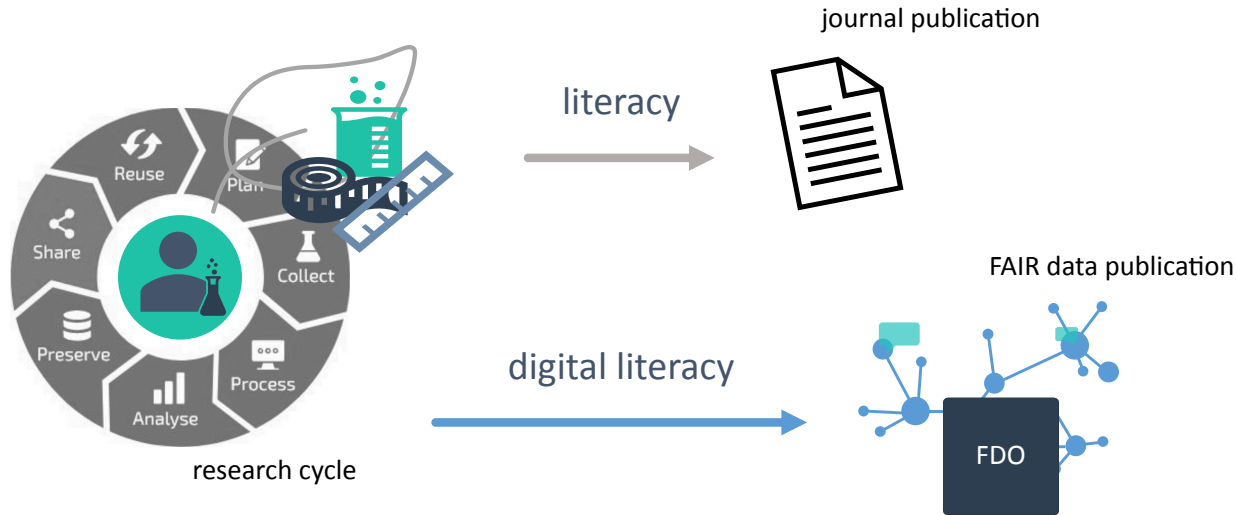
<https://nfdi4plants.org>

Driving the 'Digital Change' in science



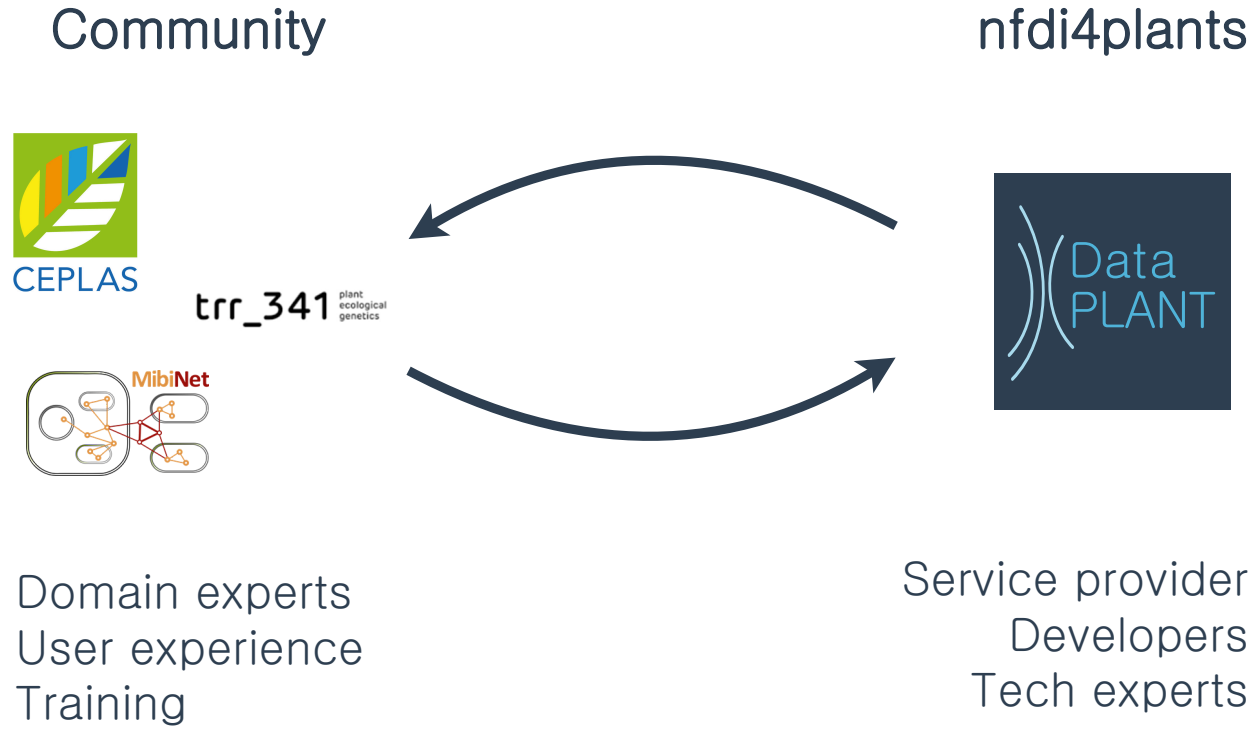
Becoming FAIR will drive science and democratization of research data

Transformation towards new competencies

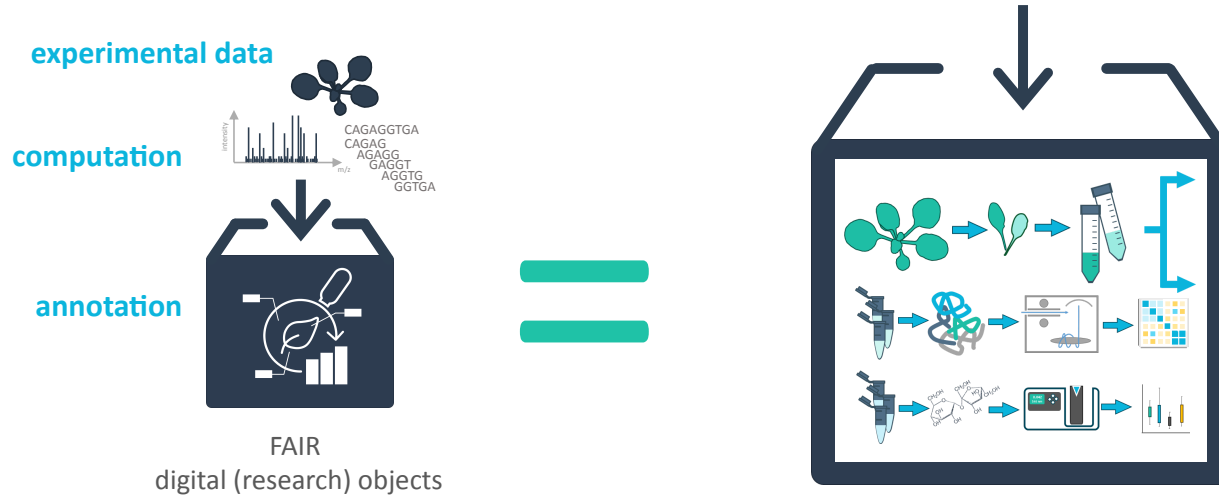


Enabling the community to participate in the RDM ecosystem

Data Stewardship for a growing community

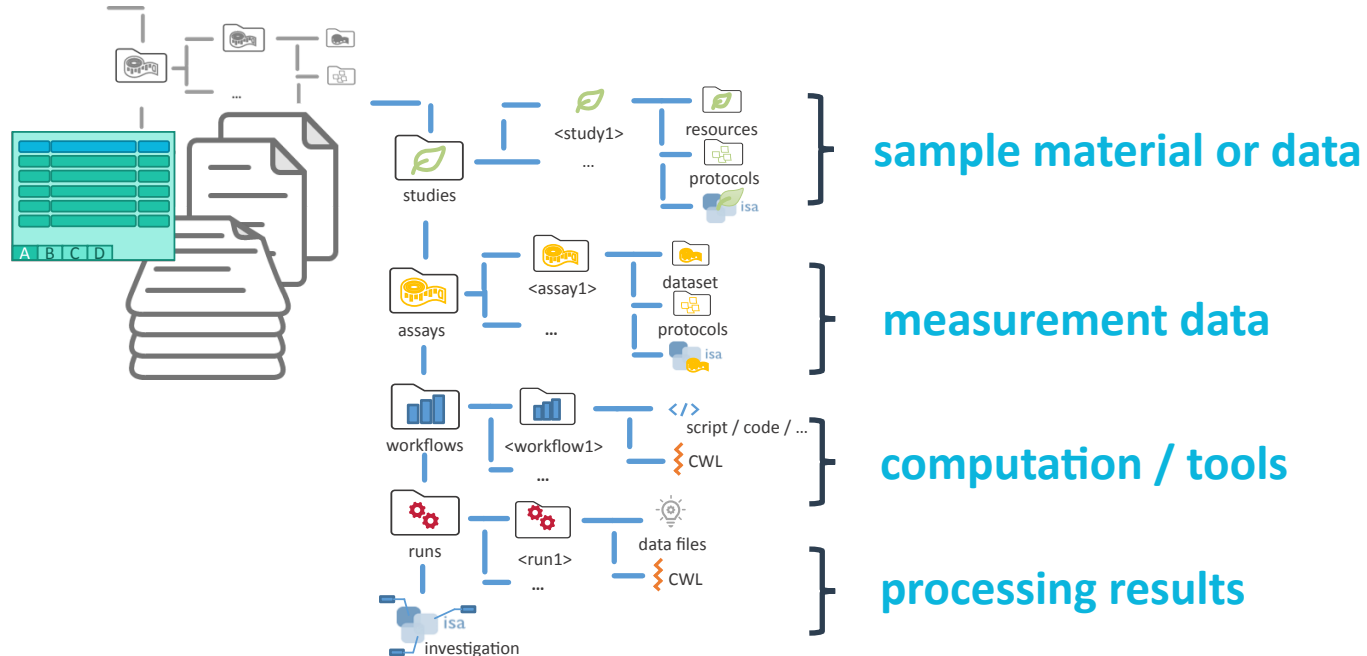


Annotated Research Context (ARC)

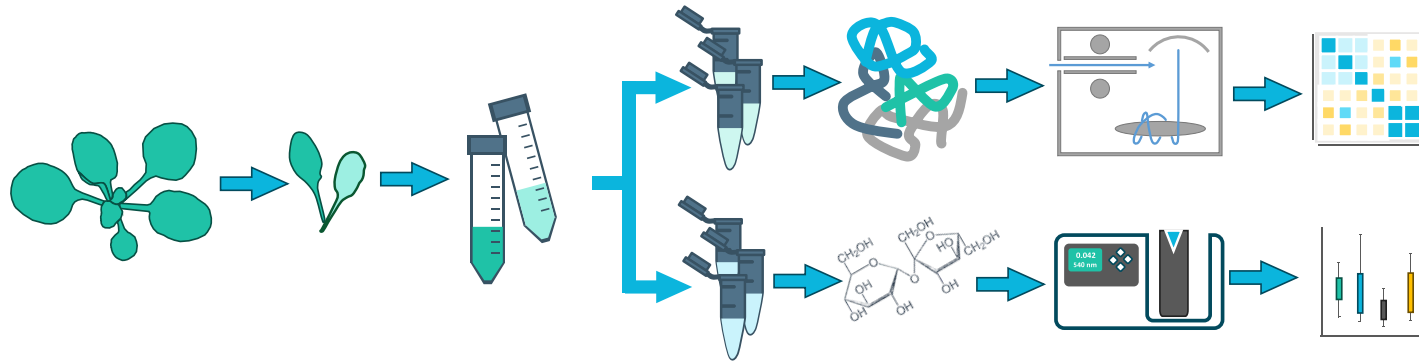


Your entire investigation in a single unified bag

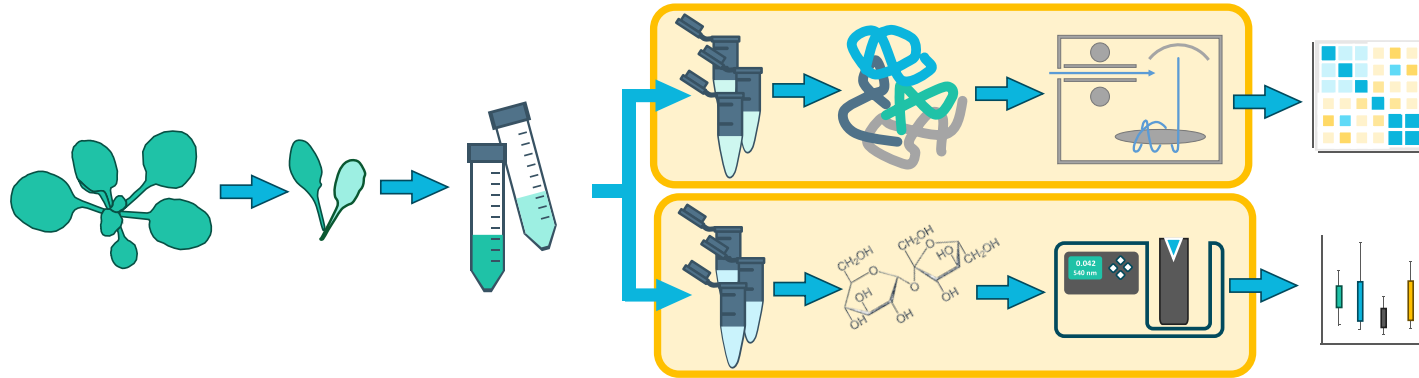
The ARC scaffold structure



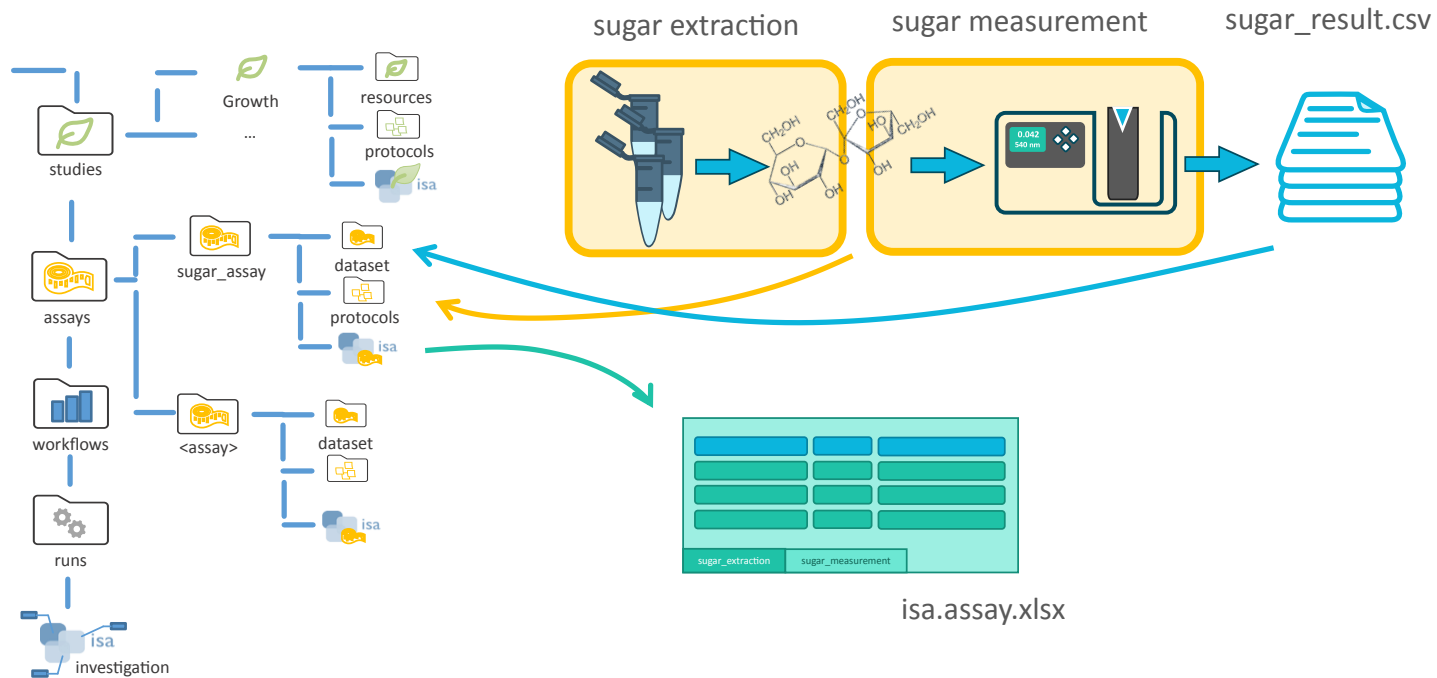
Metadata annotation – from sample to data



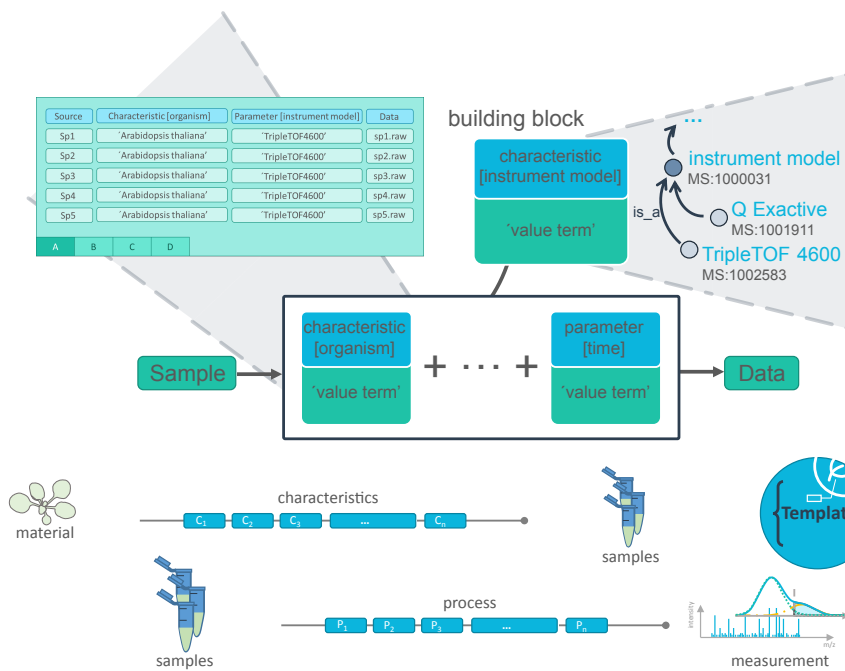
Modular separation of experimental processes



Modular separation of experimental processes

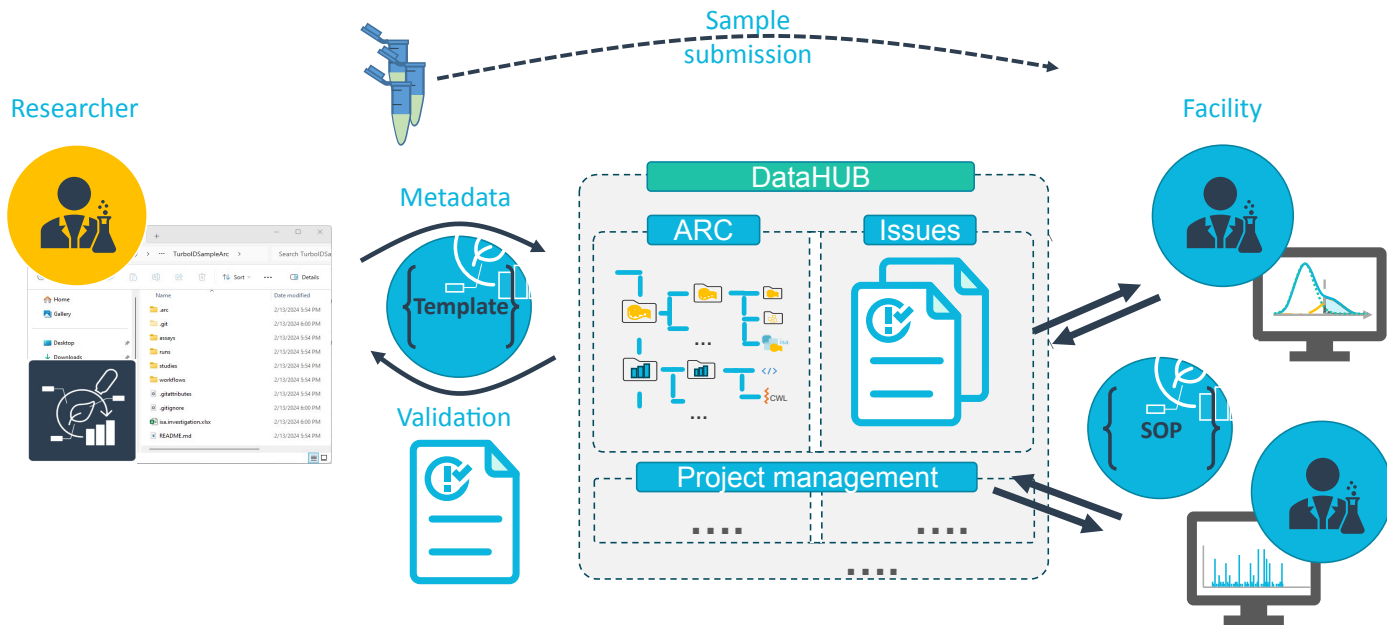


Applying standard procedures to sample records

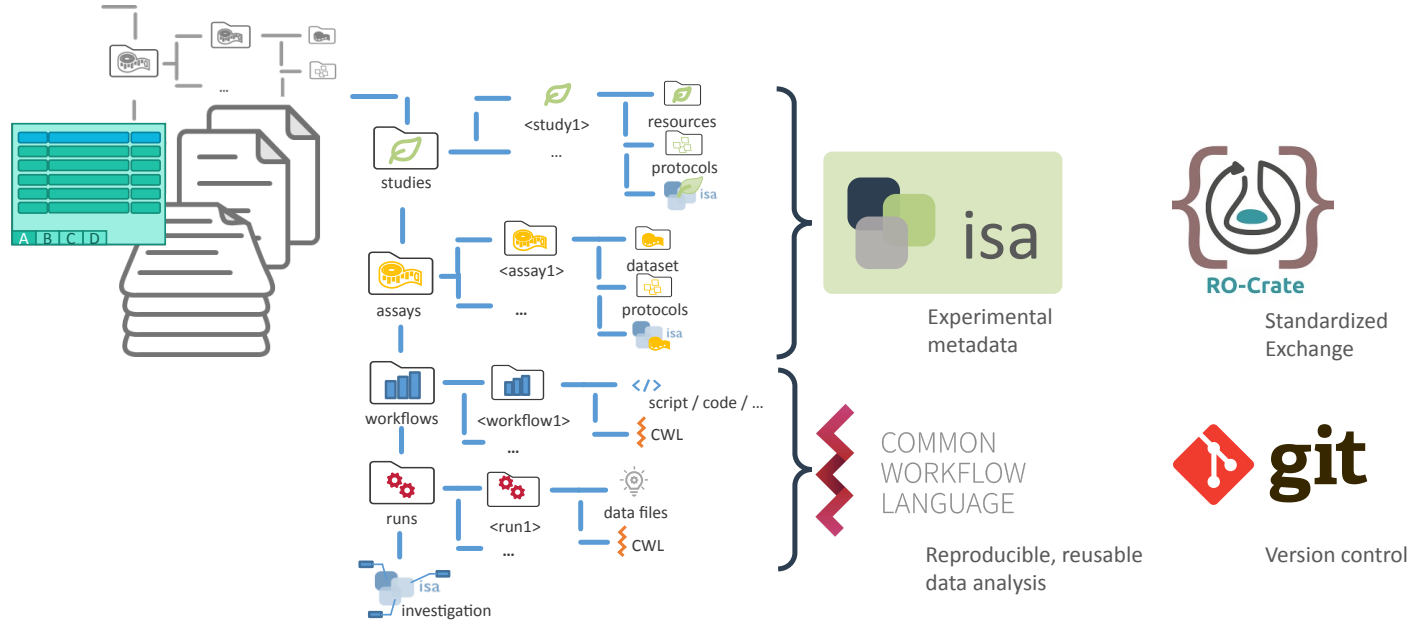


Labs and facilities can define their most common workflows as templates and SOPs

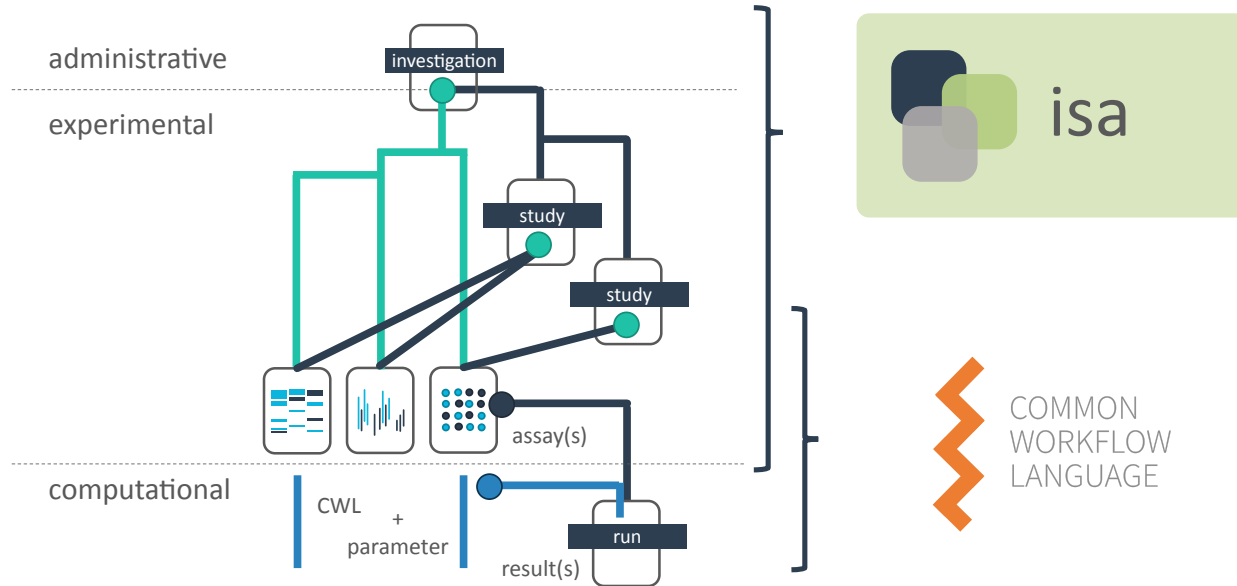
RDM to support Core Facilities



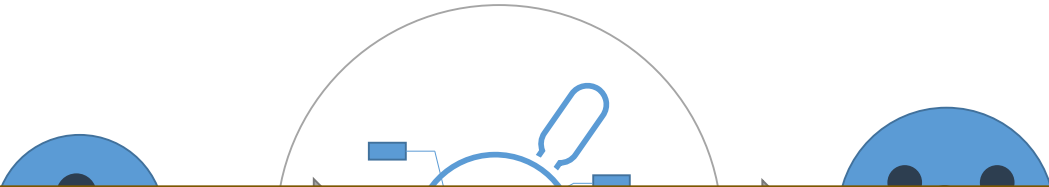
ARC builds on standards



ISA and CWL – Connected by similarity



Git is a highly flexible collaboration tool



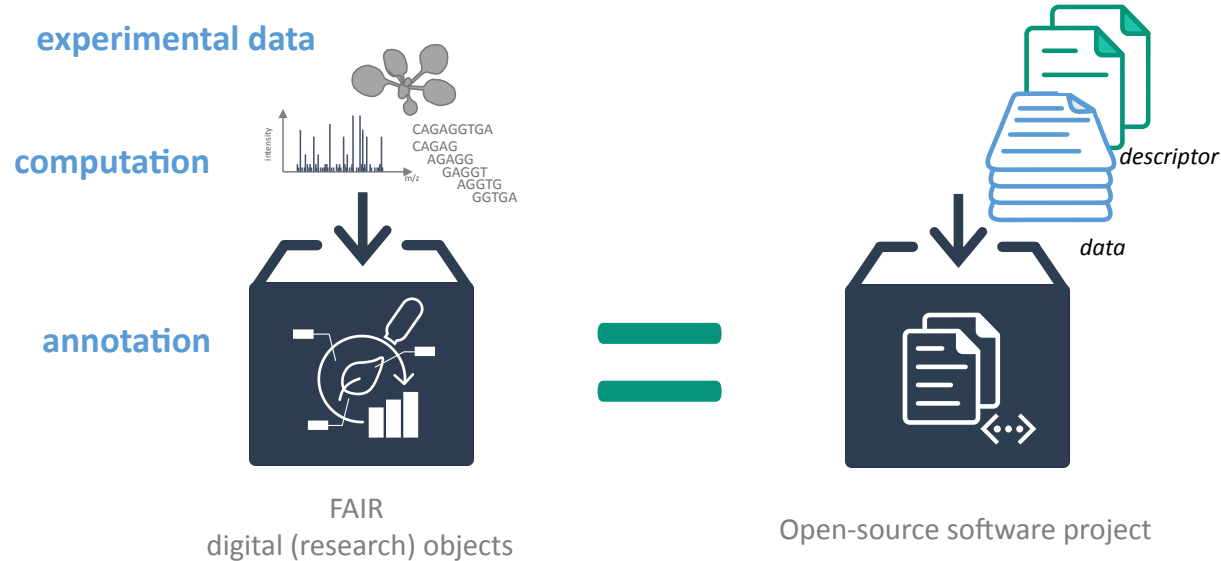
FDM-Werkstatt 2026

Does GitLab support RDM outside of standard processes? A showcase within the git.nrw project

- Session 4 | Wednesday | 09:00 AM – 12:00 PM
- Michael Werger (University of Duisburg-Essen)

- Git is an open-source Version Control System
- Fast & reliable as command line, native and web-based apps

Learning from open-source software development

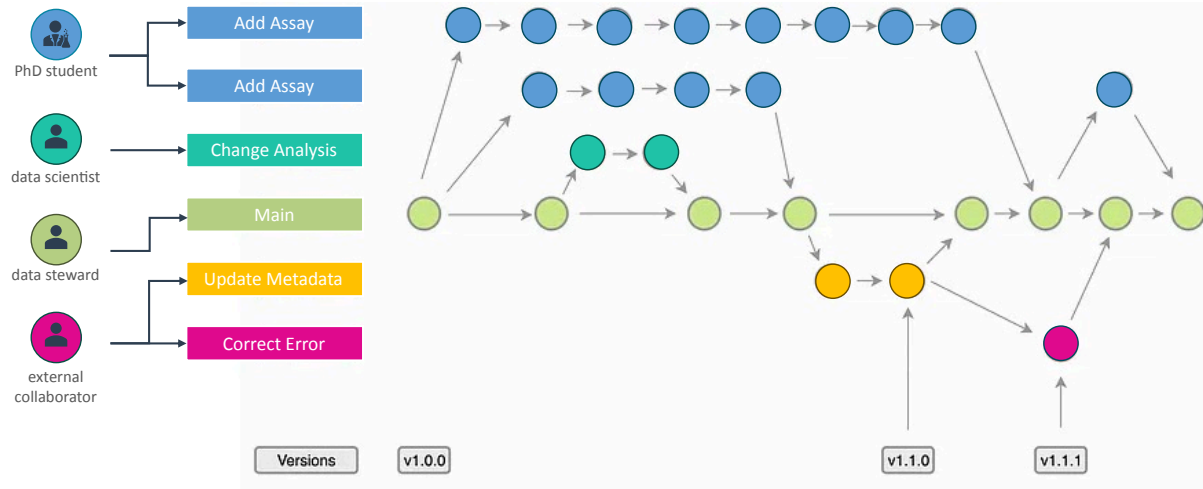


Collaborative RDM: Immutable yet evolving



Together on a stepwise path to FAIR data and democratized research

Collaborative Workflows



Domain specific conventions are easy to implement

FAIR and easy: One entry and endless possibilities



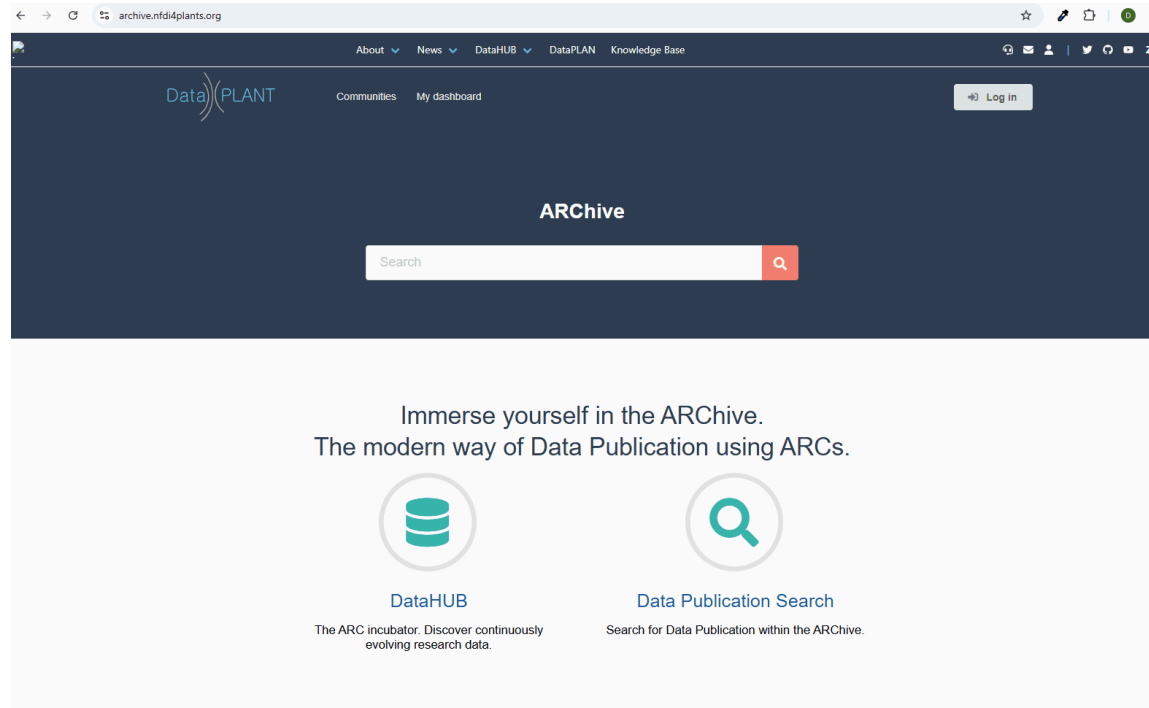
FDM-Werkstatt 2026

An Introduction to the Galaxy platform for FAIR data analysis

- Session 3 | Tuesday | 02:00 – 05:00 PM
- Saskia Hiltemann – Galaxy, DataPLANT

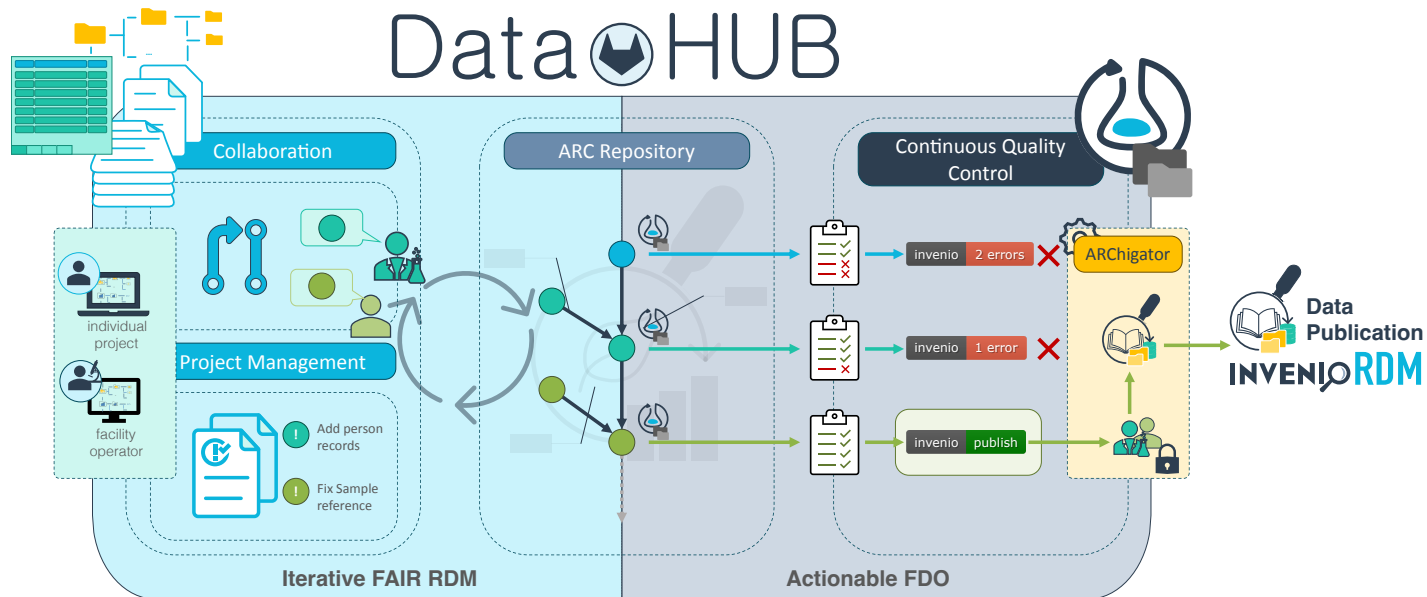
ARC enables researchers to participate in a thriving RDM ecosystem without barriers

ARChive: DataPLANT's ARC publication platform

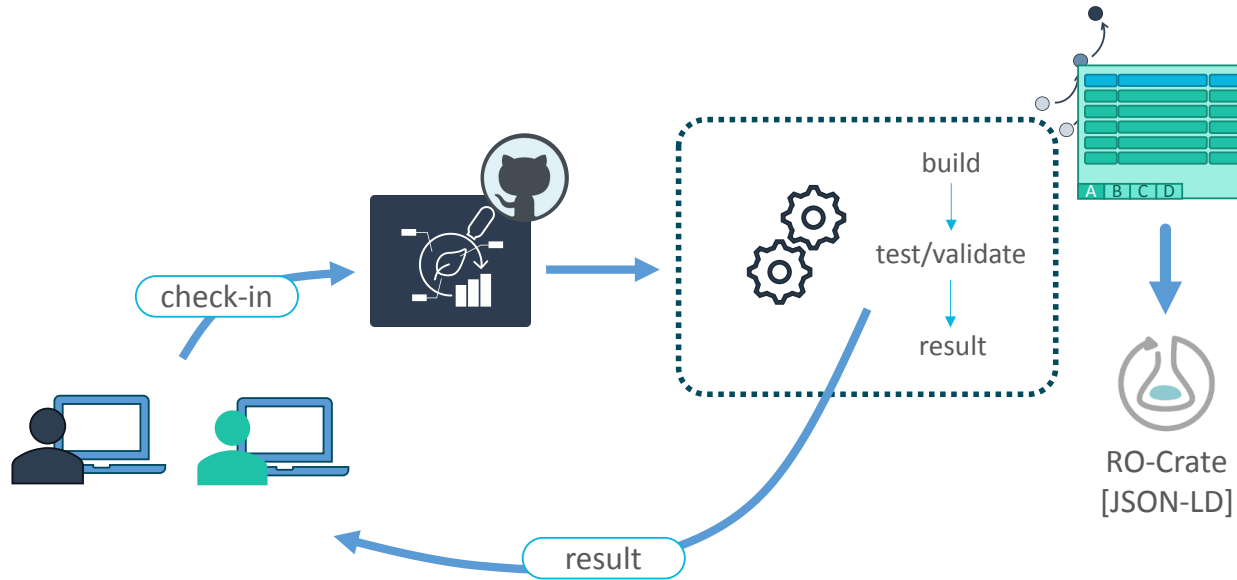


<https://archive.nfdi4plants.org>

ARC revision and publication



Automatization: “Build and validate”



Continuous integration for seamless interoperation

Validation and quality control

M

MyQuickARC

main

MyQuickARC

+

Find file

Edit

Code



add invenio package

David Zimmer authored 19 hours ago



ae9dd6a7



History

Name	Last commit	Last update
.arc	add invenio package	19 hours ago
assays	init	20 hours ago
runs	init	20 hours ago
studies	init	20 hours ago
workflows	init	20 hours ago
.gitattributes	Create ARC	20 hours ago
.gitignore	Create ARC	20 hours ago
isa.investigation.xlsx	init	20 hours ago

Project information

pipeline

passed

pipeline

passed

pipeline

passed

pipeline

passed

invenio@2.1.0

6 Critical Errors

arc_specification@2.0.0-draft

17/17

3 Commits

2 Branches

0 Tags

55 KiB Project Storage

Auto DevOps enabled

+ Add README

+ Add LICENSE

+ Add CHANGELOG

+ Add CONTRIBUTING

+ Add Kubernetes cluster

+ Add Wiki

+ Configure Integrations

Validation and quality control

The screenshot displays a GitHub Actions workflow page for the 'add invenio package' workflow. The workflow is in a 'Failed' state, indicated by a red 'x' icon and the text 'David Zimmer created pipeline for commit ae9dd6a7 19 hours ago, finished 19 hours ago'. The workflow is for the 'main' branch and has 4 jobs, with 35 seconds of execution time and 4 seconds of queue time.

The 'Tests' tab is selected, showing a summary of 10 tests, 6 failures, 0 errors, and a 40% success rate. A progress bar indicates the status of the tests. Below the summary, a table lists the test results:

Suite	Name	Filename	Status	Duration	Details
[Critical; INVESTIGATION; Investigation Title]			Failed	17.00ms	View details
[Critical; INVESTIGATION; Investigation Description]			Failed	17.00ms	View details
[Critical; INVESTIGATION; Investigation Person First Name exists]			Failed	17.00ms	View details
[Critical; INVESTIGATION; Investigation Person Last Name exists]			Failed	17.00ms	View details
[Critical; INVESTIGATION; Investigation Person Affiliation exists]			Failed	17.00ms	View details
[Critical; INVESTIGATION; Investigation Person Email exists]			Failed	0.00ms	View details

The right sidebar shows 'Project information' for the workflow, including a list of pipeline runs (all passed), commit history (3 commits), branches (2 branches), tags (0 tags), and project storage (55 KiB). It also lists configuration options like 'Auto DevOps enabled' and 'Add README'.

Validation and quality control

Ru_ChlamyHeatstress

main Ru_ChlamyHeatstress

fix headers and values for pride validation
Heinrich Luees Weil authored 9 months ago

Find file Edit Code

Project information
Algae cultures were grown mixotrophically (TAP). After 24h of 35°C/40°C the cells were shifted back to room temperature for 48h. 'omics samples were taken.

Chlamydomonas abiotic stress
proteomics + 1 more

pipeline: passed | inventory: 2.1.0 10/10
pride: 1.0.3 22/22
arc_specification: 2.0.0-draft 17/17

68 Commits
3 Branches
0 Tags

README
Auto DevOps enabled
+ Add Wiki

Created on
July 11, 2022

Name	Last commit	Last update
.arc	add pride validation package	9 months ago
assays	fix headers and values for pride validati...	9 months ago
publication	add publication, add zScores	3 years ago
runs	add ontologyenrichment workflow and ...	9 months ago
studies	fix headers and values for pride validati...	9 months ago
workflows	add ontologyenrichment workflow and ...	9 months ago
.gitattributes	rename missing samples; #2	2 years ago
.gitignore	add gitignore	2 years ago
README.md	update README.md to reinclude move...	9 months ago
isa_investigation.xlsx	fix headers and values for pride validati...	9 months ago

README.md

Validation and quality control

DataPLANT Publication Service

DATA PUBLICATION

DataPLANT

2024-12-06

<https://archive.ncf4plants.org>

INVESTIGATION

Systems-wide investigation of responses to moderate and acute high temperatures in the green alga Chlamydomonas reinhardtii.

Algae cultures were grown mixotrophically (TAP). After 24h of 35°C/40°C the cells were shifted back to room temperature for 48h. 'omics samples were taken.

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SOURCE OF PUBLICATION

Ru_ChlamyHeatstress

Ready

I agree to the [Terms & Conditions](#)

I made sure that all corresponding authors are listed

Publish

Star 2

Fork 2

Project information

Algae cultures were grown mixotrophically (TAP). After 24h of 35°C/40°C the cells were shifted back to room temperature for 48h. 'omics samples were taken.

Chlamydomonas reinhardtii

abiotic stress

proteomics

more

pipeline passed Invenio@2.1.0 10/10

pride@1.0.3 22/22

arc-specification 2.0.0-draft 17/17

68 Comments

3 Branches

0 Tasks

README

Auto DevOps enabled

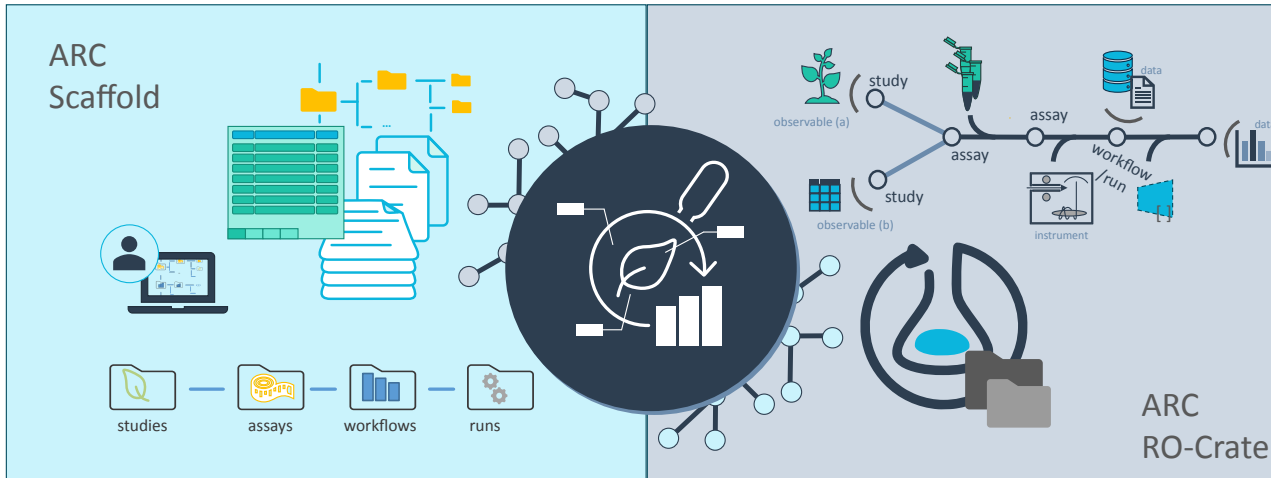
Add Wiki

Created on July 11, 2022

DataPLANT
CEPLAS

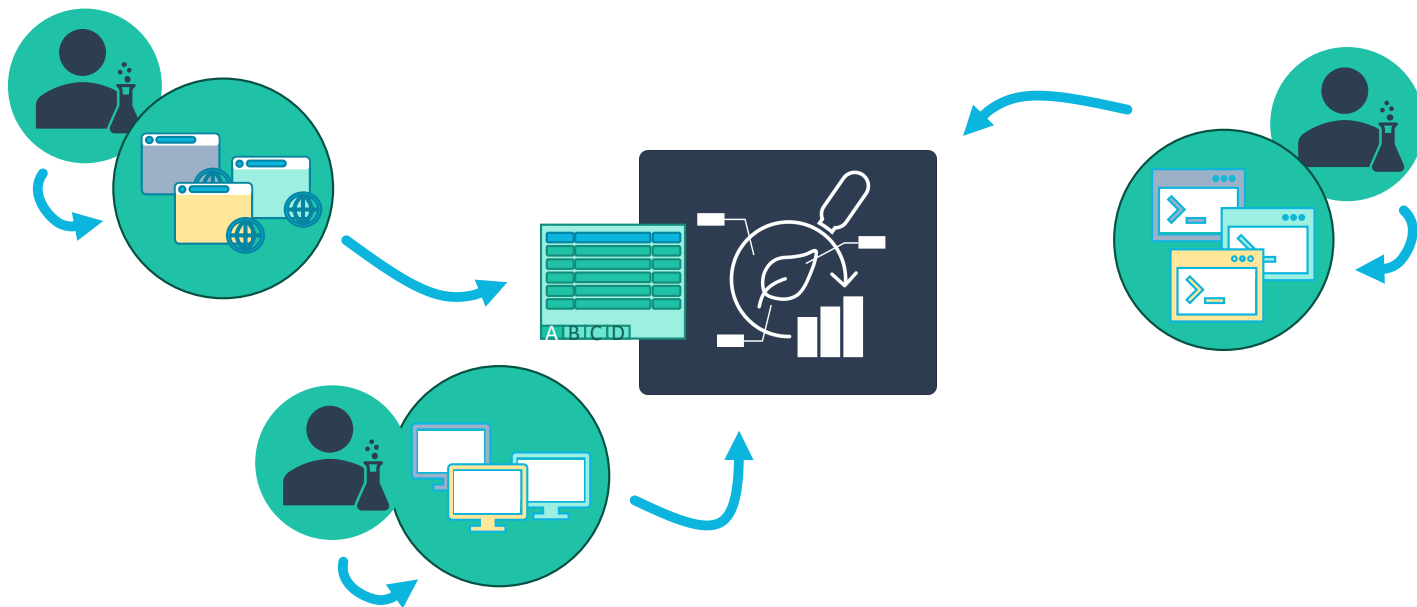
1 / 39

Two sides of the same coin



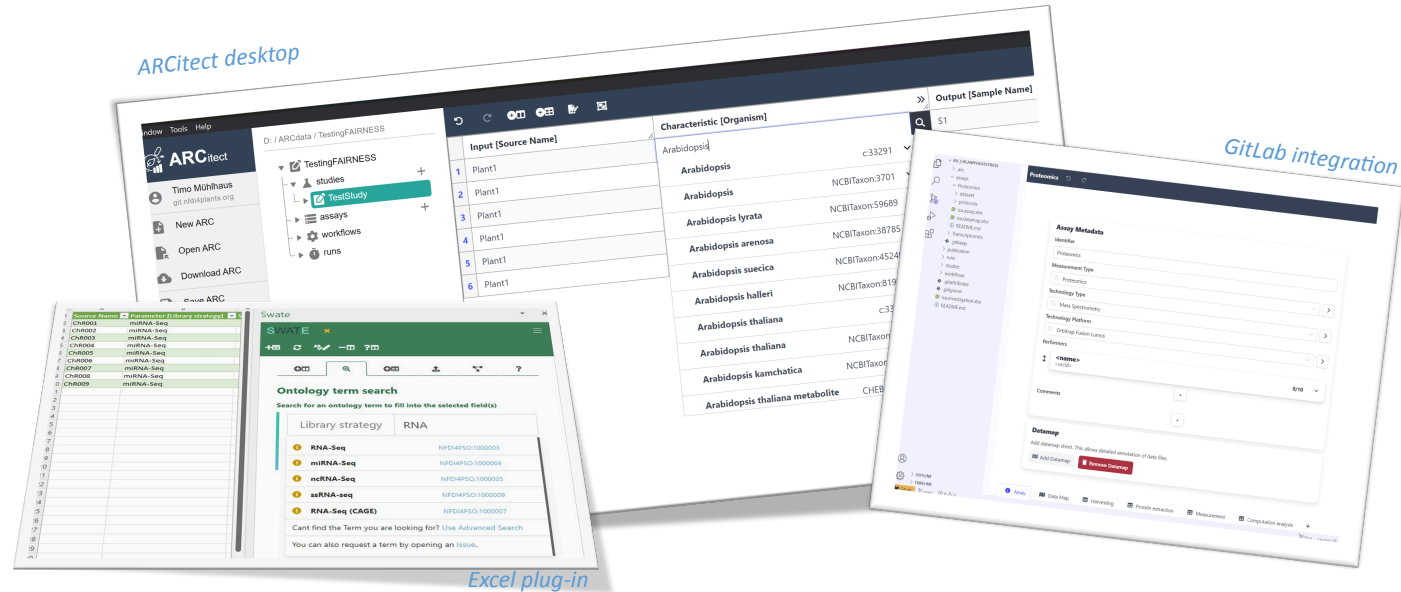
Tool development and tool assistance

Tool support



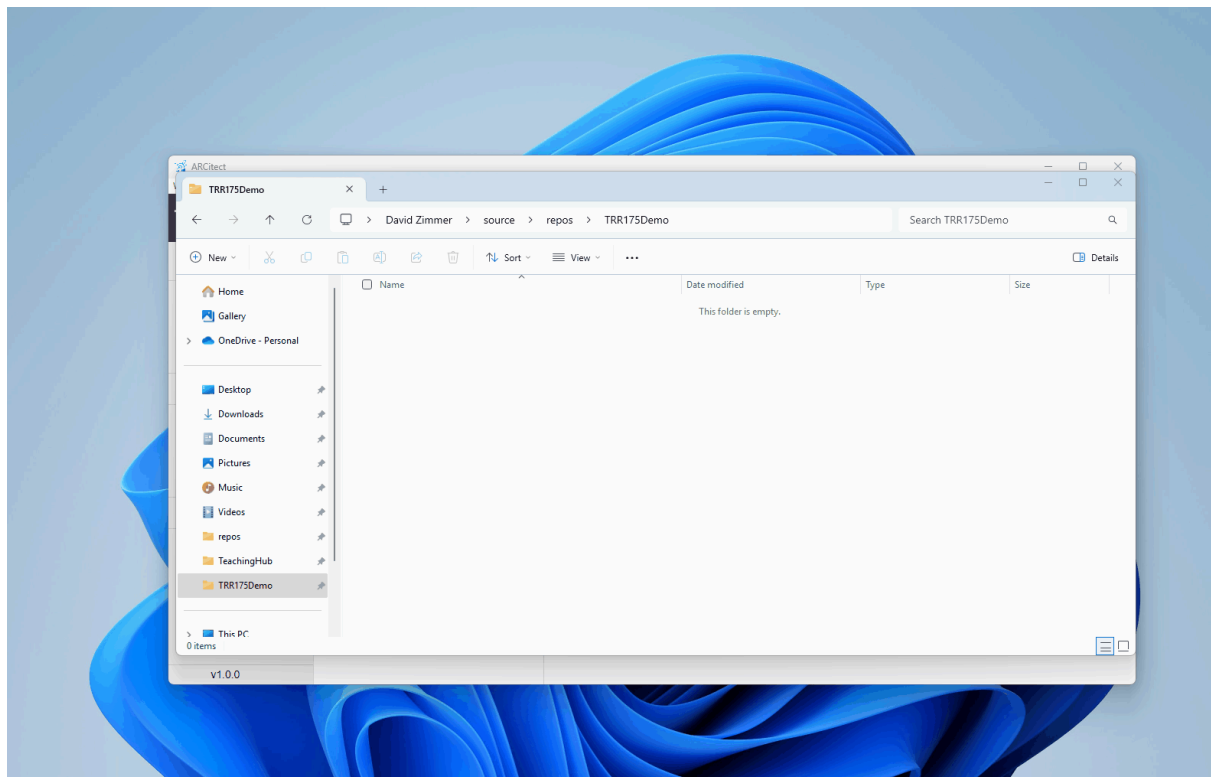
(Meta)data transparency with tool assistance but no technical lock-in

From tool development to tool assistance



Community-driven tool development for all-in-one ARC management

ARCitect





FDM-Werkstatt 2026

Turning Research Projects into FAIR Digital Objects: a Hands-On Introduction to Annotated Research Contexts (ARCs) and ELN integration

- Session 2 | Tuesday | 09:00 AM – 12:00 PM
- Sabrina Zander – MibiNet
- Yaser Alashloo – CEPLAS

Protocol:

- 1. Dissolve the supernatant.
- 2. Add 100 µl of incubation water to the panel and incubate.
- 3. Centrifuge for 10 minutes at 20,000g, 4°C.
- 4. Discard the supernatant.
- 5. Depending on the initial weighed amount, add 200 µl of freshly prepared sodium acetate (50mmol, pH 4.7).
- 6. Incubate for 20 minutes, heat (program 30).

Set up DNA digestion overnight with shaking at 37°C:

- For 100 samples: 100 x 200 µl NAC 20 mM x 20ml total volume.
- 200 µl enzymes (SMAA, KOD-RS-2) (final concentration: 10mg/ml) per 5 units per sample.
- 112 µl 4x Ampyrescense (SMAA from Amp. 10g/l) (final concentration: 40mg/ml) or 5 units per sample.

the next day:

- Phase all samples in a thermocycler for 10 minutes at 95°C to inactivate the enzymes.

Linked Resources and Experiments:

-  [https://doi.org/10.26434/chemrxiv-2024-00000](#)

Attachments:

- [arc_mibi.xlsx](#)

Workflows:

- [isa.assay.xlsx](#)

Runs:

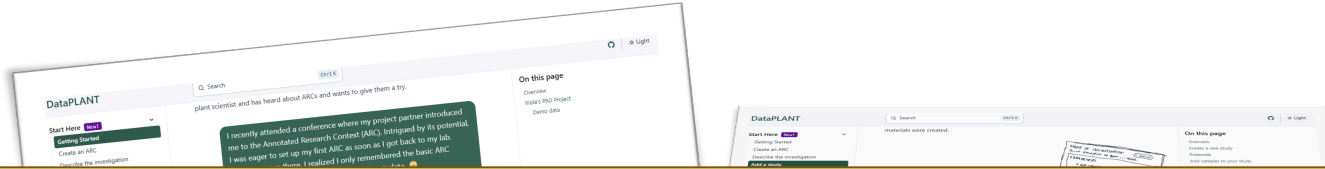
- [isa.investigation.xlsx](#)

Data Stewardship & Continuous Support

- Help Desk: <https://helpdesk.nfdi4plants.org>
- User Support chat: [#arc-user-support:matrix.org](#)
- User Support Mailing List: arc-user-support-join@lists.nfdi.de
- Events <https://nfdi4plants.org/events/>
 - Data Steward Circle
 - Infrastructure Provider Circle
 - User Support Meeting
 - Trainings and workshops



Growing specific training resources



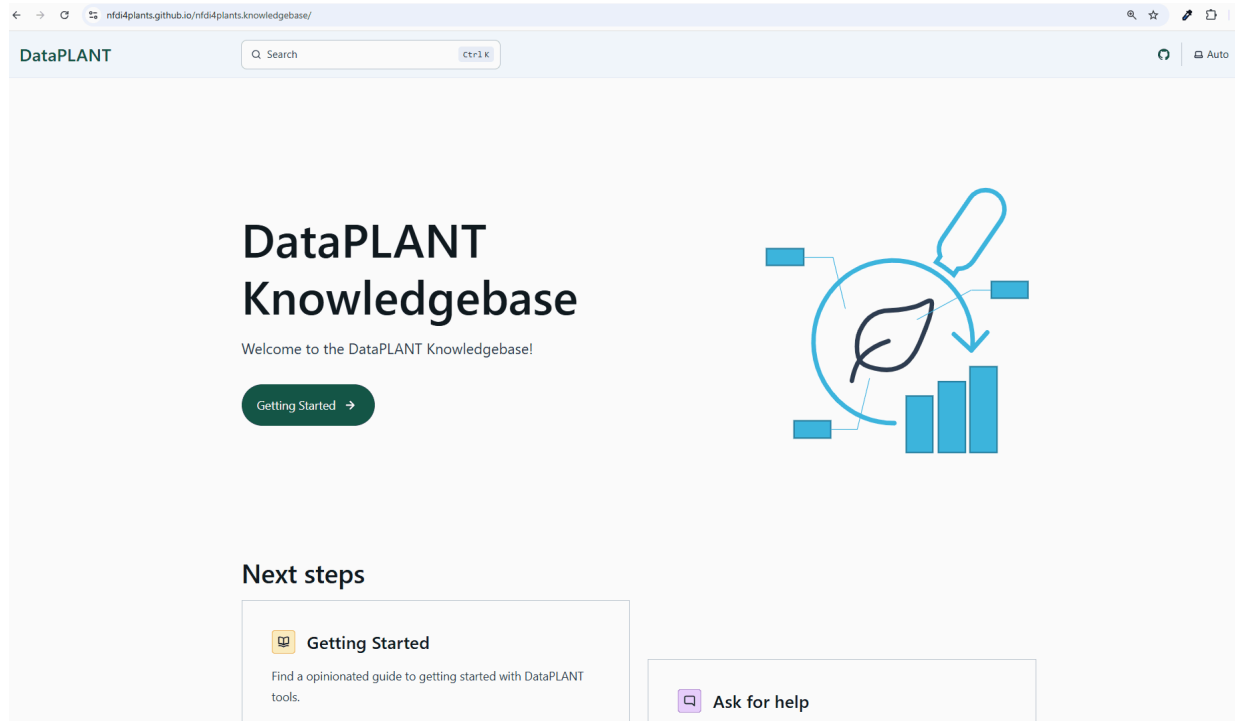
FDM-Werkstatt 2026

Supporting Research Data Management in the Life Sciences: From Challenges to Solutions with de.NBI & ELIXIR Germany

- Session 1 | Monday | 03:00 – 05:00 PM
- Helena Schnitzer (Forschungszentrum Jülich GmbH, de.NBI & ELIXIR-DE)
- Nils-Christian Lübke (Forschungszentrum Jülich GmbH, de.NBI & ELIXIR-DE)

ARC-specific training is offered independently or integrated into international training resources

Growing specific training resources



<https://nfdi4plants.org/nfdi4plants.knowledgebase/>

DMP support



The screenshot shows a web-based questionnaire titled "Practical Data Management Guide of DFG Project". The interface includes a top navigation bar with tabs for "basic info", "data type", "data volume", "data standard", and "data analysis". The main content area is divided into two columns. The left column contains the introductory text and the first section, "1. Data acquisition", which includes sub-sections for data generation, collection, and organization. The right column contains sections for user information ("1.3 Contact info of user"), data officer information ("1.4 Give the data officer name"), and additional information ("1.4 Chose some additional information"). A blue box highlights the "Single page questionnaire" text.

Livepreview

Practical Data Management Guide of **DFG Project**

This practical guide of data management in **DFG Project** should be considered as a minimum description, leaving flexibility to include additional actions of specific domain or to national or local legislation.

The practical guide of data management in **DFG Project** aims at providing a complete walkthrough for the researcher. The contents are customized based on the user input in the Data Management Plant Generator (DMPG). The practices in this guide are customized to fit related legal, ethical, standardization and funding body requirements. The suitable practices will cover all steps of a data management life-cycle:

1. Data acquisition:

- a. Data generation
Data should be generated by devices that are compatible with the open source format. The **research object** should be compliant to biodiversity protocols. The protocols used to collect phenotypic, genetic, genomic, metabolomic, RNAseq data about **research object** will be stored in a FAIR data storage.
- b. Data collection
The data collection process is conducted by experimental scientists and stewarded by **Example data officer name ABC**.
- c. Data Organization
The data organization process is conducted by **Example data officer name ABC**. The detailed organization method and procedure are reported to the PIs.

2. Annotation

1.3 Contact info of user

name of the user
Example User ABC

Email of the user
email this

1.4 Give the data officer name:
Example data officer name ABC

1.4 Chose some additional information

- ☒ The data is protected.
- ☐ Yes I will use **bioRxiv**.
- ☐ This project will be updated.
- ☐ This project has previous projects.
- ☐ This is an industry project.
- ☐ This project use **openetary softwares**.
- ☐ This project has partners.
- ☐ EU project.

Where will you submit your data as endpoints?

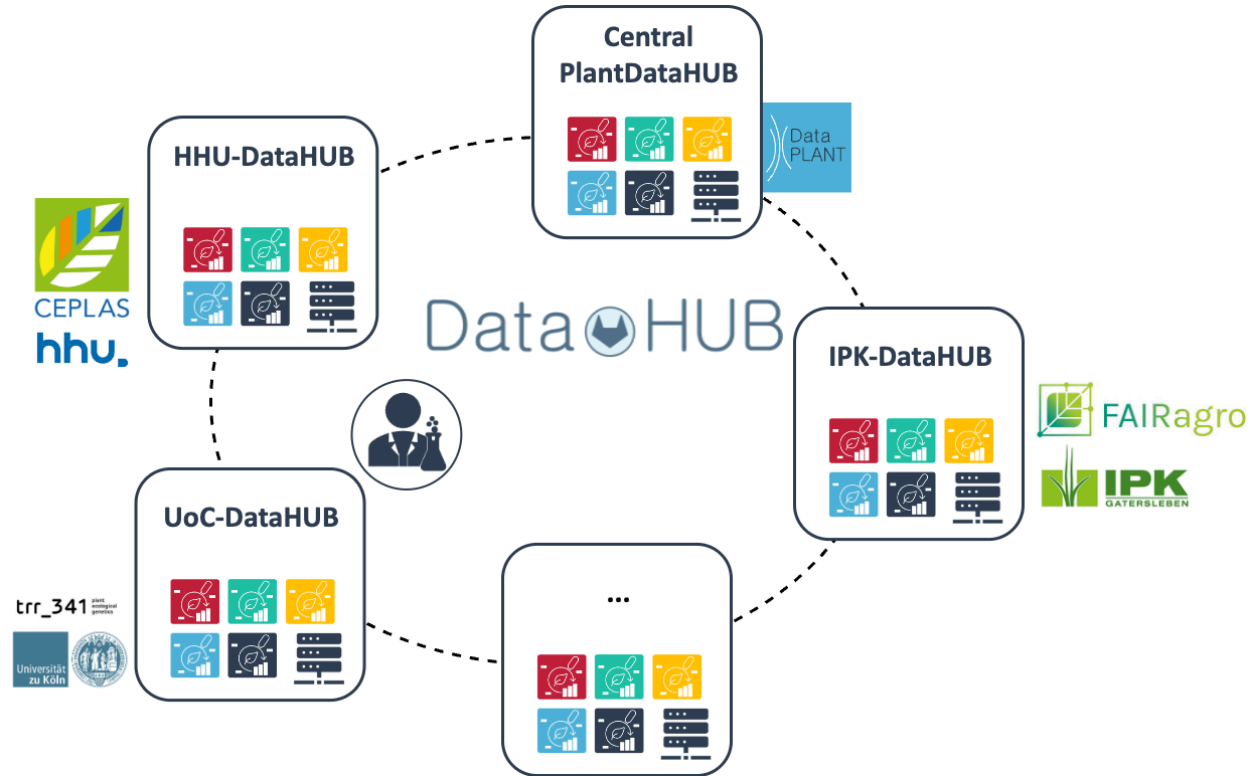
ENA, GEO, NCBI, genbank, pride, proteomexchange, zenodo, dataDryad

- ☐ ENA
- ☒ GEO
- ☒ NCBI

Single page questionnaire

Incentivizing FAIR data management

On-premise data hubs



Acknowledgements



Team Kaiserslautern

- Timo Mühlhaus
- Lukas Weil
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- Marcel Tschöpe
- Julian Weidhase

Team Jülich

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- Kathryn Dumschott
- Stella Eggels



- Björn Usadel
- Yaser Alashloo
- Vittorio Tracanna



- Sabrina Zander



- Eik Dahms